WILL ROSENBERG

ARIZONA STATE UNIVERSITY
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EDUCATION

ARIZONA STATE UNIVERSITY **Tempe, AZ**
PhD Student in Computer Science **2025 – Present**
Advisor: Yan Shoshitaishvili

WASHINGTON UNIVERSITY IN ST. LOUIS (WASHU) **St. Louis, MO**
Bachelor of Science in Computer Science, Physics, and Mathematical Sciences **2021 – 2025**
Summa Cum Laude

RESEARCH EXPERIENCE

SEFCOM LABORATORY **Tempe, AZ**
Arizona State University **Aug 2025 – Present**

- Explored program transformation to prevent bug attribution with applications in CVE dataset evaluation and data privacy.
- Implemented automated userspace program recording and fuzzing for IPC-based programs.
- Contributed to an evaluation dataset of software build processes and reviewed LLM agent logs to determine build success: "Build-Bench: Benchmarking LLM Agents on Compiling Real-World Open-Source Software" (TMLR, 2026).
- Evaluated auto-generated bug patches for root-cause alignment and patch quality; manually identified the root cause for 30+ bugs in large software projects: "Root-Cause-Driven Automated Vulnerability Repair" (arXiv preprint, 2026).

COMPUTER SECURITY AND PRIVACY LABORATORY **St. Louis, MO**
Washington University in St. Louis **Aug 2023 – May 2025**

- Applied fuzzing tools to identify potential vulnerabilities in C/C++ cyber-physical libraries.
- Categorized reported timing bugs across real-time operating systems including FreeRTOS, Zephyr, and RIOT.

QUANTUM MONTE CARLO GROUP FOR NUCLEAR PHYSICS **St. Louis, MO**
Washington University in St. Louis **May – Dec 2022**

- Used a neural network to approximate the numerical solution of a nucleon-nucleon scattering problem with the goal of applying this network in the optimization problem of nuclear interaction models.
- Presented a poster to faculty and peers from WashU and other research universities at the 2022 Midstates Consortium for Math and Science Undergraduate Research Symposium and the Washington University Physics Research Symposium.

TECHNICAL EXPERIENCE

OPEN-SOURCE CONTRIBUTIONS **2025 – Present**

- Discovered and patched a Protobuf vulnerability, earning $3,133.70 from Google's OSS VRP.
- Identified multiple Linux kernel bugs, including CVE-2025-71085 and CVE-2026-23127, and authored upstream fixes.
- Independently discovered and reported CVE-2026-23014 before public disclosure, then developed a local privilege escalation exploit and technical write-up.
- Extended Juvenal, an AI-agent orchestrator, to support dynamic workflows and verification loops for vulnerability discovery.

SHELLPHISH & BEARSHELL: CAPTURE-THE-FLAG (CTF) COMPETITIONS **Aug 2022 – Present**

- Competed in 48-hour CTF hacking competitions focusing on binary exploitation, reverse engineering, and cryptography.
- Qualified for and competed in DEF CON CTF 2026, m0leCon CTF 2026, and Securinets 2025 in-person finals with Shellphish.
- Added features and improved the reliability and speed of Shellphish's exploit-sending tool during DEF CON finals.
- Captained BEARSHELL, WashU's cybersecurity club, from 2023 to 2025.
- Created an introductory bootcamp, prepared lectures and activities for weekly meetings, and taught topics in binary exploitation, cryptography, web security, reverse engineering, and forensics.

ADDITIONAL INFORMATION

Academic Service: USENIX WOOT 2026 Artifact Evaluation Committee Member, reviewed and tested research artifacts; ACSAC 2025 CTF Organizing Committee Member, created a challenge exploring use-after-free vulnerabilities in jemalloc.

Teaching: Graduate Teaching Assistant (TA), Introduction to Cybersecurity (ASU, 2026); Head TA, Systems Software (WashU, 2024–2025); TA, Introduction to Computer Security, Object-Oriented Programming Workshop, and Introduction to Computer Science (WashU, 2021–2023).

Honors: Outstanding Senior Award (WashU Computer Science Department, 2025); Departmental Distinction (WashU Department of Physics, 2025); Inducted into Sigma Pi Sigma, National Physics and Astronomy Honor Society (2025).

Interests: Rock climbing with friends, hiking, playing chess, and solving logic puzzles.